Physics :-

(1) Convex lens

(2) A convex lens with focal length 10 cm has greater Power.

(3) Given, $u = -5 cm$
$v = 10 cm$

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{10} - \frac{1}{(-5)} = \frac{1}{f}$$

$$\frac{1}{10} + \frac{1}{5} = \frac{1}{f}$$

$$\frac{1+2}{10} = \frac{1}{f}$$

$$f = \frac{10}{3} = 3.33 cm$$

len type ⇒ Convex lens

(5)
(4) Given, $v = +50\,cm$
$m = -1$, hence image is inverted and of same size.

$$m = \frac{v}{u}$$

$$-1 = \frac{v}{u}$$

$$-u = v$$

$$-50\,cm = u$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{f} = \frac{1}{50} - \frac{1}{(-50)}$$

$$\frac{1}{f} \quad \frac{1}{50} + \frac{1}{50}$$

$$\frac{1}{f} = \frac{1+1}{50}$$

$$\frac{1}{f} = \frac{2}{50}$$

$$f = 25\,cm$$

focal length is 25 cm

$$\text{Power} = \frac{1}{f}$$

$$P = \frac{100\ (\text{in cm})}{f\ (\text{in cm})}$$

$$P = \frac{100}{25}$$

$$P = 4D$$

$$P = +4D$$